

Roll No.

TCS-501

B. TECH. (CSE) (FIFTH SEMESTER) MID SEMESTER EXAMINATION, Oct., 2023

SYSTEM SOFTWARE

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Write short notes (with the help of suitable examples) on : (CO1)

- (i) System vs. Application software
- (ii) Forward vs. Backward referencing
- (iii) Pass vs. Phase
- (iv) Compiler vs. Interpreter

OR

(b) Employ the multi-pass assembler to evaluate the following expressions : (CO1)

Expression No.	Loc	Source Statement		
1		HALFSZ	EQU	MAXLEN/2
2		MAXLEN/2	EQU	BUFEND-BUFFER
3		PREVBT	EQU	BUFFER-1
4	4034	BUFFER	RESB	4096
5	5034	BUFEND	EQU	*

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2. (a) Generate the object code for the following program : (CO1)

	PROG2	START	0
		LDT (74)	#4
		LDX (04)	#0
	MOVCH	LDCH (50)	STR1, X
		STCH (54)	STR2, X
		TIXR (B8)	T
		JLT (38)	MOVCH
	STR1	BYTE	C'GEHU
	STR2	RESB	04
		END	

OR

- (b) State the difference between SIC and SIC/XE architecture. Clearly state why SIC has 15 bits address and SIC/XE has 20 bits address field in the instruction. (CO1)
3. (a) Explain the use of data structure in pass-1 and pass-2 assemblers. What is the need for a 2-pass assembler ? Reason out with a simple example. (CO2)

OR

- (b) Consider the SIX/XE machine architecture program with only one control section, and write the object program with D and R records :

(CO2)

END4	START	0
	EXTDEF	BUFFER, BUFFEND, LENGTH
	EXTREF	RDREC, WRREC
FIRST	STL (14)	RETADR
CLOOP	+JSUB	RDREC
	(48)	
	LDA (00)	LENGTH
	CPMP (28)	#0
	JEQ (30)	ENDFIL
	+JSUB	WRREC
	(48)	
	J(3C)	CLOOP
ENDFIL	LDA (00)	=C'EOF'
	STA (0C)	BUFFER
	LDA (00)	#3
	STA (0C)	LENGTH
	+JSUB	WRREC
	(48)	
	J(3C)	@RETADR

RETADR	RESW	1
LENGTH	RESW	1
	LTORG	
BUFFER	RESB	4096
BUFFEN	EQU	*
D		
MAXLEN	EQU	BUFFEND-BUFFER
	END	FIRST

4. (a) Disassemble the following SIC object program (assume the meaningful name for the labels) opcodes : LDA-00, STA-0C, LDCH-54, STCH-54. The hex code for the ASCII character Z is 5A. (CO2)

HEXAM4 001000000014

T0010000C00100F0C100C501012541013

T00100F040000055A

E001000

OR

- (b) What is program relocation ? Explain the problems associated with it and their solutions. (CO2)
5. (a) Generate the machine code for the following by considering SIC/XE machine : (CO2)
- (i) 0000 +JSUB RDREC

(5)

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(ii) 0004 STL RETADR

(iii) 0008 LDB #LENGTH

(iv) 000A CLEAR X

Assume the opcodes are in hexadecimal :

JSUB = 48, STL = 14, LDB = 60, CLEAR = B4

The LOCCTR value for RDREC = 1036

RETADR = 0030, LENGTH = 0033

OR

(b) Find out the target address for the following object codes, by considering SIC/XE machine architecture :

(CO2)

(i) 032600

(ii) 010060

(iii) 022030

content of X = 000090, B = 006000, PC = 004000

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B. TECH. (CSE) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

OPERATING SYSTEM

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) A shared variable x , initialized to zero, is operated on by four concurrent processes A, B, C, D as follows. Each of the processes A and B reads x from memory, increments by one, stores it to memory, and then terminates. Each of the processes C and D reads x from memory, decrements by two, stores it to memory, and then terminates. Each process before reading x invokes the P operation (i.e., wait) on a counting semaphore S and invokes the V operation (i.e., signal) on the semaphore S after storing x to memory. Semaphore S is initialized to two. Calculate the maximum possible value of x after all processes completes their execution, also show the calculations.. (CO2)

OR

- (b) A certain computation generates two arrays p and q such that $p[i] = f(i)$ for $0 \leq i < n$ and $q[i] = g(p[i])$ for $0 \leq i < n$. Suppose this computation

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is decomposed into two concurrent processes A and B such that A computes the array p and B computes the array q. The processes employ two binary semaphores X and Y, both initialized to zero. The array p is shared by the two processes. The structures of the processes are shown below :

Process P1: private i; for (i=0; i<n; i++) { p[i] = f(i); ExitP1 (X, Y); }	Process P2: private i; for (i=0; i<n; i++) { EntryP2 (X, Y); q[i] = g (p[i]); }
--	---

Write the correct implementation of ExitP1 and ExitP2, so that both the processes are synchronised. (CO2)

2. (a) What is a race condition in critical section ? Explain how a critical section avoids this condition. Explain all conditions that satisfy the solution to critical section problem. (CO2)

OR

- (b) Describe the differences among short-term, medium-term, and long-term schedulers. Also explain which of them is most frequently invoked. (CO2)

3. (a) Calculate the following for the given set of processes shown in table below using round robin scheduling algorithm with time quanta 3 ms

and context switch time as 1 ms. (Assume that context switch time at the end of all processes is negligible) : (CO2)

- (i) Avg Waiting Time
- (ii) CPU Utilisation
- (iii) Throughput

Process	Burst Time (ms)	Arrival time
P1	10	2
P2	12	5
P3	8	1
P4	14	3

OR

- (b) Calculate the following for the given set of processes shown in table below using Shortest remaining time scheduling algorithm and context switch time as 1 ms. (Assume that context switch time at the end of all processes is negligible) : (CO2)

- (i) Avg Turnaround Time
- (ii) CPU Utilisation
- (iii) Avg Response Time

Process	Burst Time (ms)	Arrival time
P1	8	5
P2	10	3
P3	5	0
P4	7	7

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4. (a) What are system calls ? Explain different types of system calls with example. Also, differentiate between system calls and function calls.

(CO5)

OR

- (b) Draw and explain in details seven state diagram of a process cycle in an operating system.

(CO1)

5. (a) What are different types of Operating Systems ? Explain at least *five* types. Which type of O. S. is suitable in an interactive environment ?

(CO1)

OR

- (b) What is a process and a thread ? Explain different types of thread. What are the differences between process and thread ?

(CO1)

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B. TECH. (CSE) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
DATABASE MANAGEMENT SYSTEMS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Discuss main characteristics of database approach. How does it differ from file system ? (CO1)

OR

- (b) What are the software components in a client server DBMS ? Compare the two-tier and three-tier client server architecture. (CO1)
2. (a) Differentiate between total participation and partial participation with suitable example. (CO2)

OR

- (b) Describe weak entity set with example. What is the role of partial key in weak entity set ? (CO2)
3. (a) Explain the three-level schema architecture. How are the different schema layer related to the physical and logical data independence ?

(CO1)

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(2)

OR

- (b) Which relational algebra operations require the participating tables to be union compatible ? Give the reason in detail. (CO3)

4. (a) What are the different types of Data Models in DBMS ? Explain them. (CO1)

OR

- (b) Consider the following bank's description :

Bank have Customer. Banks are identified by a name, code, address of main office. Banks have branches. Branches are identified by a branch_no., branch_name, address. Customers are identified by name, cust-id, phone number, address. Customer can have one or more accounts. Accounts are identified by account_no., acc_type, balance. Customer can avail loans. Loans are identified by loan_id, loan_type and amount. Account and loans are related to bank's branch.

Draw an E/R diagram that represents the above information. (CO2)

5. (a) "Each super_key is a superset of some candidate key." Justify the statements with suitable examples.

Identify super key, candidate key and primary key for the following relation : (CO2)

Name	Age	City
Ravi	24	Deharadun
Divya	23	Delhi
Ravi	22	Delhi
Dev	23	Delhi

(3)

OR

- (b) Consider the below schema and write the following queries in SQL and relational algebra :

CUSTOMER (account_no, Customer_name, Customer_city)

LOAN (loanno, Branch_name, amount)

DEPOSITOR (Customer_name, account_no)

BORROWER (customer_name, loanno)

- (i) Find all tuples of the relation customer who lives in the city 'Dehradun'.
- (ii) Find those customer names who have either a loan or account in the bank.
- (iii) Find the names of all the customers who have loan at the 'SBI-PATEL NAGAR' branch.
- (iv) Find the all the loan numbers of 'SBI- PATEL NAGAR' branch where the amount is more than ₹ 8,000.
- (v) Find those customer names who have loan of amount ₹ 6,000 in the bank.

(CO3)

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B. TECH. (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

COMPUTER BASED NUMERICAL AND STATISTICAL TECHNIQUE

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Using Newton-Raphson method, Find the root of the equation $x \sin x + \cos x = 0$ up to three decimal places. (CO1)

OR

- (b) Find a real root of the equation $\cos x = 3x - 1$ correct to three decimal places using Iteration method. (CO1)

2. (a) Using Bisection method, compute the root of the equation $e^x - 3x = 0$ correct to three decimal places between 1 and 2. Also prove its rate of convergence is linear. (CO1)

OR

- (b) Define Polynomial. Using Lagrange's interpolation formula, find $f(9)$ from the following table : (CO2)

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(2)

x	$f(x)$
5	150
7	392
11	1452
13	2366
17	5202

3. (a) Marks obtained by the candidates in an examination are given as under : (CO2)

Marks Obtained	No. of Students
10—20	42
20—30	45
30—40	39
40—50	48
50—60	19

Find the number of candidates who obtained less than 44 marks.

OR

- (b) Solve the system of linear equations, using Gauss-Seidel's method performing five iterations : (CO1)

$$x + y + 54z = 110$$

$$27x + 6y - z = 85$$

$$6x + 15y + 2z = 72$$

4. (a) Using Newton's divided difference formula, find $f(x)$ from the following table : (CO2)

x	$f(x)$
0.5	1.625
1.5	5.875
3.0	31.0
5.0	131.0
6.5	282.125
8.0	521.0

OR

- (b) Find the number of terms required in the expansion of e^x such that their sum gives the correct value up to decimal 10 places at $x = 1$. (CO1)
5. (a) Use Gauss' forward interpolation formula to find the sales by a concern for the year 1936 given : (CO2)

x	Sales (in thousand)
1901	12
1911	15
1921	20
1931	27
1941	39
1951	52

(4)

OR

- (b) From the following table, estimate the number of students who obtained marks between 40 and 45 : (CO2)

Marks	No. of Students
30—40	31
40—50	42
50—60	51
60—70	35
70—80	31

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B. TECH. (CSE) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

MACHINE LEARNING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Rohith Sharma is the captain of the Indian cricket team. The runs scored by him in the last six matches are as follows :

12, 46, 34, 12, 51, 32.

Calculate the following statistical analysis on runs scored by Rohith Sharma : (CO1)

- (i) Mean, Median, Mode and Range of scores
- (ii) Absolute Deviation of scores
- (iii) Average Deviation of scores
- (iv) Standard Deviation of scores
- (v) Total Sum of Squares of scores

P. T. O.

(2)

OR

(b) The government introduces a new scheme named "Gruha Jyoti" for the public that aims to raise the living conditions of residents. As per the new scheme, the electricity charges are as follows : (CO1)

- (i) Up to 150 units ₹ 0
- (ii) 151-200 units ₹ 5 per unit
- (iii) 200-250 units ₹ 10 per unit
- (iv) Above 250 units, ₹ 12.5 per unit

In the last 6 months, electricity consumption in units at Ajay and Vijay houses has been 225, 175, 320, 280, 190, 149 and 320, 290, 380, 210, 225, 187. The above-mentioned values are stored in the Python List data structure.

Develop a menu driven Python program to perform the following operations :

- (i) To calculate the total electricity charges for the last six months of electricity consumption at Ajay's house and print the amount.
- (ii) To calculate the total electricity charges for the last six months of electricity consumption at Vijay's house and print the amount.
- (iii) To find the maximum electricity consumption value in last six months at Ajay and Vijay's houses.
- (iv) to find the minimum electricity consumption value in last six months at Ajay and Vijay's houses.

2. (a) Apply the K-means clustering algorithm to assign the given data into two clusters and assume the initial centroids are (2,1) and (2,3) : (CO5)

Objects	O1	O2	O3	O4	O5	O6
X	1	2	2	3	4	5
Y	2	1	3	2	3	5

OR

- (b) Apply DBSCAN clustering algorithm to the given data to form clusters with epsilon = 3.5 and Minimum Points = 3 : (CO5)

Objects	O1	O2	O3	O4	O5	O6	O7	O8
X	5	8	3	4	3	6	6	5
Y	7	4	3	4	7	7	1	5

3. (a) Discuss K-means clustering along with its advantages and disadvantages. Explain working of Silhouette Scores to find the value of K. (CO2)

OR

- (b) Discuss Agglomerative hierarchical clustering and its different methods to measure for the distance between clusters. (CO2)
4. (a) Compare the following : (CO3)
- (i) Supervised Machine Learning vs. Unsupervised Machine Learning
 - (ii) Python List vs. Tuple

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OR

(b) Find the inverse of a matrix :

(CO3)

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 2 & 9 \end{pmatrix}$$

5. (a) Apply Apriori Algorithm on the given Transaction Data-set and Generate Rules. Consider the values as Support = 50% and Confidence = 75% :

(CO6)

Transaction ID	Items Purchased
1	Bread, Cheese, Egg, Juice
2	Bread, Cheese, Juice
3	Bread, Milk, Yoghurt
4	Bread, Juice, Milk
5	Cheese, Juice, Milk

OR

(b) Write short notes on the following :

(CO6)

(i) Principal Component Analysis

(ii) Backward Feature Elimination and Forward Feature Selection

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B. TECH. (CSE SPECIALIZATION) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

USER INTERFACE DESIGN

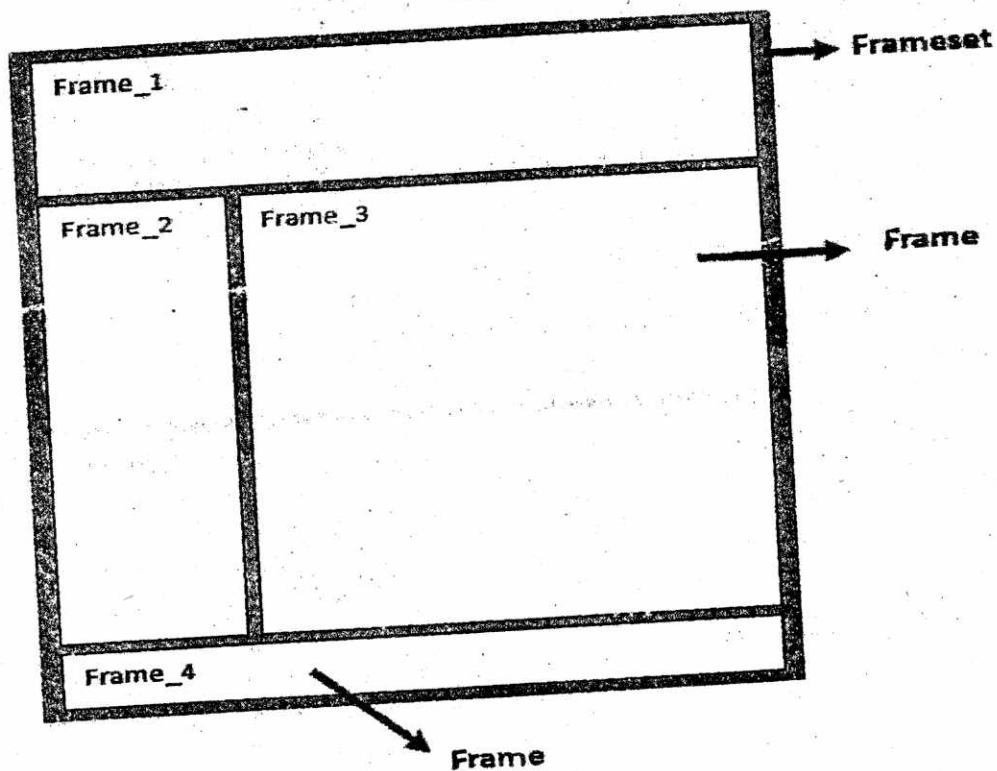
Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Write HTML code for designing the specified frame.

(CO2)

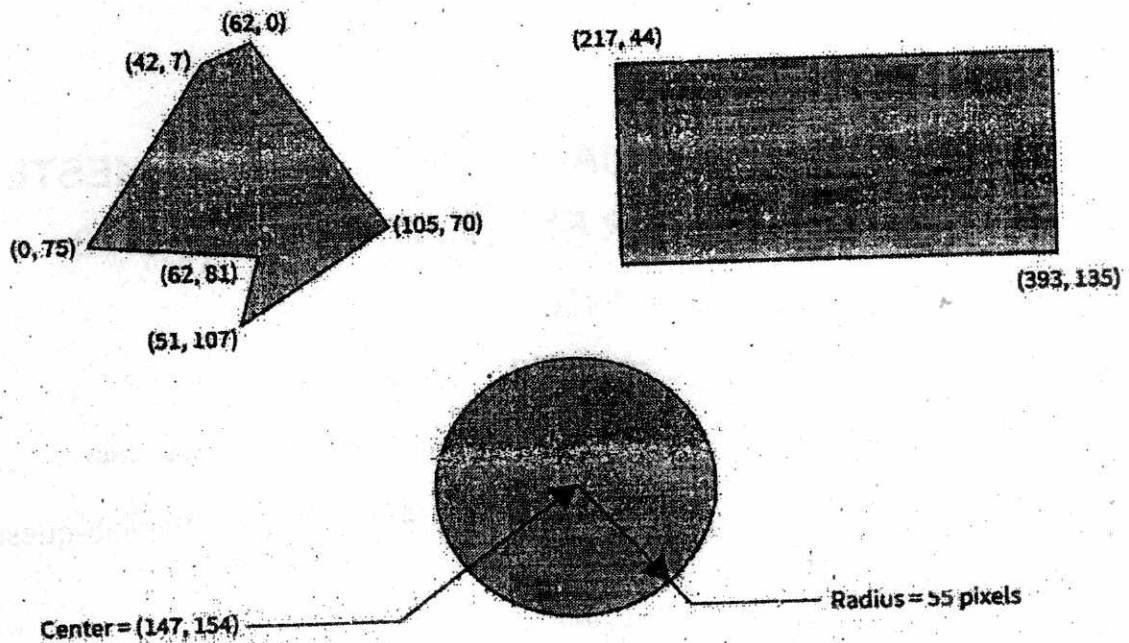


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- (b) Create the necessary code to build a map. Use a single picture to combine a circle, rectangle, and polygon. (CO2)



2. (a) Describe the problems with color selection for interface design. Why is internationalization necessary? Talk about the relevant problems. (CO2)

OR

- (b) What is the proper way to use the following HTML5 semantic elements?

“header,” “article,” “section,” and “footer”? (CO2)

3. (a) When building a UI, briefly define the BOX model in the cascading styles sheet. List the different window management techniques. (CO2)

OR

- (b) What does HTML5's two-web storage mean? What are the disadvantages of cookies, and how would you integrate audio and video in HTML5? (CO2)

4. (a) Create the HTML code necessary to design the provided form.

Name	
Address	
Password	
Job Experience	
D.O.B.	dd- - - - yy yy ☐
Email	
Search	
Month	----- ☐
Week	week --, -- ☐
Time	--:-- -- ☐
Tel:	
URL	
Color	<input type="color"/>
Range	<input type="range"/>
File	Choose File No file chosen
Qualifications	10th ☐ 12th ☐ Graduation ☐
Category	GEN ☐ OBC ☐ SC ☐ Other ☐
	submit

- (b) Name a few HTML5 formatting components. SVG HTML5 Graphics
What are they ?
5. (a) Describe the features of GUI. GUI and web interface design
compared and contrasted.

OR

- (b) What distinguishes usability from user experience, and how ? How
you make sure that everyone, including people with impairment
utilize your designs ?

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TCS-523

B. TECH. (CSE) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
BIG DATA AND MACHINE LEARNING USING PYTHON

Time : 1½ Hours

Maximum Marks : 50

Note : (i) All questions are compulsory.

(ii) Answer any *one* of the sub-questions among (a) and (b) and from main question.

(ii) Total marks for each sub-question carries is 10 marks.

1. (a) Evaluate various Quality Metrics for any ML model. (CO1)

OR

(b) Sam found how many hours of sunshine vs. how many icecreams were sold at the shop from Monday to Friday : (CO1)

No. of Hours	Ice-Creams
2	4
3	5
5	7
7	10
9	15

Predict the number of icecreams when the sunshine hours is 6 hours.

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(2)

(CO3)

2. (a) Difference between the following :

- (i) Regression and Correlation
- (ii) Naïve Bayes and Decision Trees.

OR

(b) Compare Deep Learning and Machine Learning Algorithms taking their usability, capability, algorithm design approach, accuracy, output, performance as necessary parameters. (CO2)

3. (a) Using KNN Algorithm, Predict the Class where Brightness is 20 and Saturation is 35, taking $K = 5$ using below data : (CO1)

Brightness	Saturation	Class
40	20	Red
50	50	Blue
60	90	Blue
10	25	Red
70	70	Blue
25	80	Blue

OR

(b) Compare Big Data Analytic and Traditional Analytics System. (CO2)

4. (a) Analyze and Compare different types of possible Data in real world. (CO2)

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(CO3)

(3)

OR

(b) What is the benefit of using below data structures in Python ? Mention usage of at least 10 useful predefined functions of each : (CO2)

(i) Data Frames

(ii) Tuples

5. (a) Elaborate the type of relationship lines that can exist between different attributes in a given dataset. (CO3)

OR

(b) What are IQRs ? How the outliers can be handled via varied approaches ? (CO2)

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TCS-527

B. TECH. (CSE) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

APPLIED MACHINE LEARNING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) List out and briefly explain the unsupervised algorithms. (CO1)

OR

- (b) Explain the feature engineering methods with example. (CO1)

2. (a) What is the feature in machine learning ? Why feature engineering is important for machine learning ? What is feature extraction in machine learning techniques ? How to select the feature in machine learning algorithms ? (CO1)

OR

- (b) (i) Explain in stages the data management life cycle.

- (ii) Explain the types of Big Data with an example. (CO1)

3. (a) Explain the process of obtaining the principle components and its relevance in feature reduction. (CO1)

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(2)

OR

- (b) (i) What is the role of regression model in exploratory data analysis ?
(ii) What is the principle of ordinary least square in linear regression ?
What are the pros and cons of regression models ? (CO2)

4. (a) (i) What is precision-recall curves ?
(ii) What is F-score ?
(iii) Explain a simple precision-recall curve and the corresponding ROC curve for skewed data. (CO2)

OR

- (b) (i) How does the polynomial regression model work ?
(ii) How is logistic regression model different from linear regression ? (CO2)
5. (a) What is difference between overfitting and underfitting ? Explain with an example. (CO2)

OR

- (b) Explain the Ridge regression and Lasso regression with a numerical example. (CO2)

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TCS-549

B. TECH. (CSE) (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

DATA SCIENCE VISUALIZATION WITH TABLEAU SPECIALIZATION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Describe how data visualization can be used to visualize historical data, such as population demographics, migration patterns, or trade routes, to gain insights into past civilizations. (CO1)

OR

- (b) Analyze the classic case of the London 1855 cholera epidemic and explain how data visualization was pivotal in changing the face of public health and epidemiology. (CO2)

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(2)

2. (a) Imagine you are working on a data analysis project involving numerical and textual data. How would you apply the "Different Data Calls for Different Views" principle to select the most suitable visualization methods for each data type, and how might this choice impact the project's outcomes and insights ? (CO2)

OR

- (b) Imagine you are leading a data visualization project for a financial institution, and the primary objective is to help clients understand and track their investment portfolios. How would you define the visualization design objectives to ensure clarity, provide real-time updates, and enhance the user experience for investors with varying financial expertise ? (CO3)
3. (a) You are a data scientist working on a research project in the field of climate science. You must visualize temperature, precipitation, and atmospheric data collected over several decades. Explain how you would select different data visualization techniques to effectively communicate climate trends, extreme weather events, and atmospheric patterns to scientists, policymakers, and the general public, each with varying expertise in the field. (CO2)

(3)

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OR

- (b) You are leading a team tasked with creating a data visualization for a national election campaign. Each team member is wearing one of the "eight hats" of data visualization design. How would you ensure effective collaboration among these diverse roles to produce a visualization that informs voters, conveys the candidate's message, and respects data privacy and transparency ?
- (CO3)

4. (a) Imagine you are a financial analyst in a rapidly changing market environment. Your role involves analyzing financial data from various sources, including stock prices, economic indicators, and news sentiment. Explain how you would use data analysis techniques to identify trends, correlations, and potential investment opportunities, considering the need for timely decision-making in a volatile market.

(CO3)

OR

- (b) You are a data analyst for a healthcare organization, and you have been tasked with analyzing patient medical records to identify trends in treatment outcomes. Describe the steps to prepare and familiarize yourself with the data.

(CO2)

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5. (a) You work for a financial institution, and your team analyzes transaction data from customer accounts.' Describe how you would take into account the physical properties of this economic data, such as data types (e.g., transaction amounts, dates), scales (e.g., monthly or yearly aggregates), and formats (e.g., structured tables or unstructured text comments), to perform fraud detection and financial trend analysis effectively. (CO2)

OR

- (b) You are a data visualization specialist working for a medical research institute. Your task is to create a data visualization that effectively communicates the findings of a complex clinical trial involving multiple variables, outcomes, and patient demographics. How would you apply data visualization techniques to ensure that healthcare professionals, and researchers can easily comprehend the trial results and make informed decisions regarding treatment ? (CO3)

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B. TECH. (CSE) (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

CLOUD BASED APPLICATION DEVELOPMENT AND MANAGEMENT

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Healthcare systems around the world are unsustainable. The global pandemic has only exacerbated existing staffing and physician shortages, increased cost pressures, and increased service backlog and capacity issues. For global virtual care company, Teladoc Health, the partnership between HITRUST and Microsoft enables data security, ensures regulatory compliance, and helps their platform maintain and grow trust with partners and patients. Evaluate Cloud Data centers from perspective of above mentioned organization. (CO1)

OR

- (b) When Provincial Health Services Authority (PHSA) began seeing the need to deliver faster, more comprehensive health insights to healthcare workers across British Columbia, it took action. Old, unstandardized requests were slowing data analysis and the sharing of insights. To

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standardize and speed data access, PHSA moved its healthcare analytics platform to Microsoft Azure, creating a readily accessible data analysis platform that is scalable and secure. Data requests that once took many months now take only days. Because of the sensitive nature of health data, PHSA has worked closely with Microsoft to make sure health data never leaves Canada and is kept highly secure. Evaluate Software defined networks and storage from perspective of above mentioned organization. (CO2)

2. (a) Hamad Medical Corporation (HMC) is the leading healthcare provider in Qatar. It was looking for a security information and event management (SIEM) solution. "Our vision was to deliver security without compromising access and usability. We also wanted to ensure patient and corporate confidentiality at all times," says Richard Storey, Executive Director/Acting Chief of Health Information and Communication Technology at HMC. With Microsoft Sentinel, HMC gained 360-degree, real-time visibility over cybersecurity, making proactive responses faster for supporting staff to access information. Analyze SLA from perspective of above mentioned organization. (CO2)

OR

- (b) Growing healthcare organization MultiCare Health System teamed up with Microsoft and Microsoft partner 3Cloud to meet changing demands in health data access prompted by the use of fitness trackers and phones. MultiCare wanted a scalable, interoperable data exchange framework to connect providers, patients, and payers-key for success in population health and value-based care. It also required an adaptable

(3)

solution. Working with Microsoft FHIR experts and partnering with 3Cloud, MultiCare built a unique, fully managed solution using Microsoft Azure. The results have surpassed expectations : in three months, MultiCare has seen a significant reduction in administrative burden with a \$50 per-chart cost savings and a 65 percent reduction in labor. Analyze the AWS Compute Services from the perspective of above mentioned organization. (CO3)

3. (a) Mercy, one of the largest healthcare systems in the United States, aided its digital transformation by using Microsoft Azure to decrease operational costs, improve the patient experience, and give its providers better access to data. The company wanted to modernize its infrastructure and extract more value from its sizeable data archives to keep up with evolving patient and doctor expectations. With support from Microsoft, Mercy identified business outcomes and undertook a digital transformation to achieve them. On Azure, the company improved efficiency, connected previously siloed data, reduced patient stay durations, and built a foundation for ongoing innovation. Analyze AWS Storage Services from the perspective of above mentioned organization. (CO2)

OR

- (b) When Centro Hospitalar de Trás-os-Montes e Alto Douro needed a way to protect sensitive, highly confidential patient data and applications in its hybrid environment, it chose Microsoft Security solutions. The hospital system's future-forward strategy calls for a dynamic, cloud-

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based security approach that will stay on top of threats and position the organization for cloud migration. It now has a centralized view of security events with Microsoft Sentinel, increasing frontline defense with Azure Automation and Microsoft Defender for Cloud. With Azure Arc extend Centro Hospitala Azure platform protections to its on-premises assets, the hospital system considers its IT security vaccinations up to date. Analyze AWS communication services from the perspective of above mentioned organization. (CO3)

4. (a) CooperSurgical, a leading fertility company, adopted Microsoft Dynamics 365 as a unified CRM solution to deliver transparency during a time of rapid global growth. The company nearly doubled in size in two years, making it difficult to get an accurate overview of operations and opportunities. Many recent acquisitions were using their own CRM and ERP systems, and knowledge sharing was a time-consuming process. CooperSurgical decided to adopt Microsoft solutions globally to enhance forecasting, relationship management, and productivity. Now, the company is poised to bring healthcare products and services to patients more quickly than ever. Analyze Google app engine architecture from the perspective of above mentioned organization. (CO3)

OR

- (b) The Mount Sinai Health System (Mount Sinai), an integrated medical system, is innovating at the leading edge of medical care using AI solutions on Microsoft Azure. Since launching its Clinical Data Science

(5)

(CDS) initiative in 2015, Mount Sinai has improved care through analytics and the creation of tools that predict the needs of patients. In 2021, it migrated the CDS infrastructure to Azure to rescale compute capacity and better meet current and future needs. Through its strategic partnership with Microsoft, Mount Sinai is accelerating its use of AI innovation to identify risks to patients, including malnutrition, delirium, and falls. Now, hospital stays have been reduced, mortality rates are falling, and providers are engaged. Analyze Microsoft azure concepts from the perspective of above mentioned organization. (CO2)

5. (a) San Raffaele University and Research Hospital is one of the largest of its kind in Italy, as well as a pioneer of cutting-edge translational research and technology adoption at both a national and international level. During the pandemic, the organization saw an opportunity to start using artificial intelligence to better triage incoming symptomatic patients and identify their future reaction to the virus. Fast forward three years, the pilot project has been turned into an intelligent data solution powered by AI – which San Raffaele is going to use to transform clinical research, improve the end-to-end patient journey inside the hospital and accelerate its own move towards precision medicine. Analyze the green cloud architecture from the perspective of above mentioned organization. (CO2)

OR

- (b) Health First, a fully integrated delivery network in Florida, created a data management solution by joining its existing data warehouse

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(6)

architecture with Azure Synapse Analytics. While collecting information from more than 60 applications, Health First faced challenges in scaling its on-premises data warehouse. Additionally, the health network wanted to explore AI, predictive analytics, and other innovative technologies while maintaining a high level of security for patient data. It deployed Azure Synapse Analytics to scale up its data integration and storage capabilities in a secure cloud system. By creating this system, Health First lowered costs, improved workload processing times, and offered better data access for providers. Analyze Active Directory Federation Services (ADFS) from the perspective of above mentioned organization.

(CO3)

Roll No.

TCS-571

B. TECH. (CSE) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

BIG DATA VISUALIZATION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) You are a data visualization specialist working for a financial institution. Your task is to create and adapt visualizations to represent complex financial data sets, including stock market trends, investment portfolios, and economic indicators. How would you approach the design and adaptation of these visualizations to emphasize key financial concepts and enable clients to make informed investment decisions in a rapidly changing market environment ? (CO1)

OR

- (b) You are tasked with visualizing sales data for a retail company. What visualization techniques would you choose to help the company understand sales trends and make informed decisions ? (CO2)

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(2)

2. (a) As an educator, you are teaching a statistics course to a group of undergraduate students. Describe how you would use techniques for visual data representations to help your students grasp complex statistical concepts, such as probability distributions and hypothesis testing, through visual aids and interactive tools, ensuring that they develop a deeper understanding of the subject matter. (CO2)

OR

- (b) Imagine you are working on a data visualization project for a global health organization tasked with presenting COVID-19 vaccination rates across different regions. How would you use visual analysis to uncover meaningful patterns and disparities in the data? (CO3)

3. (a) You work for a data-driven startup company that specializes in financial analytics. Your team is tasked with creating a data visualization dashboard for investors. Describe how you would address the key factors in this project. (CO2)

OR

- (b) You work for a news organization covering a complex environmental issue, such as deforestation in a tropical rainforest. Explain how each of the "eight hats" of data visualization design would play a crucial role in developing a visualization that informs the public, influence policymakers, and sheds light on the environmental impact while considering the ethical implications of data sourcing and presentation. (CO3)

(3)

4. (a) You are a data analyst working for a meteorological agency, and you are tasked with analyzing weather data from various sensors located across a region. How would you consider the physical properties of this data, such as data type (e.g., temperature, humidity), scale (e.g., daily or hourly readings), and format (e.g., numerical or textual), to ensure accurate analysis and visualization of weather patterns? (CO3)

OR

- (b) You work for an environmental organization tasked with visualizing a complex dataset that tracks various environmental factors, such as temperature, pollution levels, and wildlife populations, across multiple regions and over several decades. How would you apply data visualization techniques to create a compelling and informative visualization that encourages public awareness and policy changes regarding environmental conservation? (CO2)

5. (a) You are tasked with refining the editorial focus for a data visualization project on urban development. How would you ensure that the focus aligns with the project's goals and meets the needs of urban planners and policymakers? (CO2)

OR

- (b) Imagine you are a financial analyst in a rapidly changing market environment. Your role involves analyzing extensive financial data from various sources, including stock prices, economic indicators, and news sentiment. Explain how you would use data analysis techniques to identify trends, correlations, and potential investment opportunities, considering the need for timely decision-making in a volatile market. (CO3)

Roll No.

TCS-572

B. TECH. (CSE) (AI & ML) (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

APPLIED MACHINE LEARNING USING PYTHON

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-question.

(ii) Each question carries 10 marks.

1. (a) Draw and explain multi-way if-elif-else statements. List some of the features of Python. (CO1)

OR

- (b) Give a comparison between lists, tuples, dictionaries and sets with examples. (CO1)

2. (a) Define Machine Learning. Describe the steps in designing learning system.

OR

- (b) Differentiate between Training error and Testing error. Explain Linear Regression with example. (CO1)

3. (a) Differentiate between supervised and unsupervised training. Explain with suitable examples. (CO2)

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(2)

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OR

- (b) Write a Python program to draw a scatter plot comparing two subject marks of Mathematics and Science. Use marks of 10 students. (CO2)

Test Data :

math_marks = [88, 92, 80, 89, 100, 80, 60, 100, 80, 34]

science_marks = [35, 79, 79, 48, 100, 88, 32, 45, 20, 30]

marks_range = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]

4. (a) What is feature engineering ? Why is it essential ? Explain the methods of feature engineering. (CO2)

OR

- (b) What are the different methods for measuring classifier performance ? (CO2)

5. (a) Consider the following training set having three different attributes namely x_1, x_2, x_3 : (CO1)

X_1	X_2	X_3	Class
-1	1	1	Negative
5	1	0	Positive
0	2	1	Negative
-4	-1	0	Negative
0	0	1	Positive
9	2	1	Positive
2	2	0	Negative

(3)

Using the K-nearest algorithm (kNN) algorithm with $K = 3$, classify the following points :

(i) (1, 1, 2)

(ii) (2, 1, 1)

(iii) (0, 0, 0)

OR

(b) Explain the following :

(i) Dimension Reduction

(ii) Gradient Descent

(iii) Lasso and Ridge Regression

Roll No.

TCS-591

B. TECH. (CSE) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
COMPUTER SYSTEM SECURITY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Describe dirty copy-on-write (COW) vulnerability. How it has been exploited. What were the adverse impacts of that exploitation ? (CO1)

OR

- (b) Describe the following security appliances : (CO1)
Firewall and intrusion detection system.

2. (a) Demonstrate the working mechanism of the buffer overflow attack using a programming example. (CO2)

OR

- (b) Demonstrate the working mechanism of the return-to-libc attack. How this attack can be mitigated ? (CO2)

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(2)

3. (a) Investigate the procedure of fuzzing using a figure. How would you compare the mutation-based fuzzer and generation-based fuzzer ?

(CO4)

OR

- (b) Discuss a few methods which can be used to prevent the buffer overflow attack.

(CO4)

4. (a) Demonstrate the procedure of the format string vulnerability and attack with the help of a program.

(CO3)

OR

- (b) What is a robust application software from a security point of view ? Explain different tools and techniques that can be used for writing the robust application software.

(CO3)

5. (a) What is the use of authentication protocols ? How would you compare the 2-factor user authentication protocol and the 3-factor user authentication protocol ?

(CO5)

OR

- (b) The sensitive data from the server's database can be revealed through the steps of the heartbleed attack. Justify this statement. Use an example in your answer.

(CO5)